STAFF REVIEW · H446 197 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. 9.

2. 11111111.

3. 4.

4. 5, the last item pushed.

I Do — source worked steps

1. Predict — The newest value 5 is on top, so Pop removes 5.

2. Confirm and explain — The stack remains [3,8] from bottom to top.

We Do — guided CHECK

1. Confirm and explain
Remove A: front=1, count=2. Insert D at slot0: rear=1, count=3. Physical D,B,C differs from logical B,C,D; modulo reuses the freed slot.

You Do — source mark schemes

1. A remains. The pops remove C then B.

2. Physical slots [40,20,30]; front=1, rear=1, count=3. Logical order is 20,30,40. Reject enqueue when count==3 and dequeue when count==0; increment indexes modulo 3.

3. The most recent change must be undone first, which follows LIFO; FIFO would undo oldest changes first.

Exit Ticket — source answer

1. False: a queue is FIFO, removing the earliest inserted item.